Business 35801
Corporation Finance for Executives
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Snap Case

Questions for Write-Up and Class Discussion

Read the case and answer the following questions.

1. Exhibit 10 reports the annual projected free cash flows from Morgan Stanley. Reconcile their formula

   FCF = Adj EBITDA – Cash Taxes – CapX - Change in NWC – Stock Based compensation

   with our regular formula

   *FCF= EBIT * (1-tax) + Depreciation –CapX – Change in NWC*

   (You need to reconcile the formula, and not the actual calculations. You may not have all information to do that).

Hint: EBIT = EBITDA – Depreciation; Adjusted EBITDA = EBITDA + Stock Based Compensation.

2. What is the NPV of the projected FCFs (and terminal value) assuming the firm was entirely equity financed (under Morgan Stanley assumptions)? What discount rate is appropriate?

Hint #1: Asking to compute NPV of an "entirely equity financed firm" is equivalent to asking to compute NPV of OA.

Hint #2: Recall the valuation would be as of March 2017. To simplify, take the valuation time as of January 2017

Hint #3: What is the discount rate to discount just the OA (as this question asks)?

3. What is the NPV of project FCF (and terminal value) using the Weighted Average Cost of Capital (WACC) approach, assuming the firm maintains a constant 2% debt-to-equity (D/V) ratio in perpetuity, as per Morgan Stanley assumptions? Given your answer, compute Snap's stock price/share. How does it compare with the value provided in the case? Perform a "sensitivity analysis" to the assumptions made.

Hint #1: The case has an approximation for WACC. Please, compute your own value.

Hint #2: Snap has $2 billion in excess cash (see Exhibit 10). Equity = NPV(FCFs) + Excess Cash.

Hint #3: In a sensitivity analysis, build a matrix with different combinations of reasonable r_{WACC} and growth rates g.

4. What is the NPV of project FCF (and terminal value) using the Adjusted Present Value (APV) approach, assuming the Snap raises $1,000,000,000 of debt soon after the IPO and keeps the level of debt constant in perpetuity (assume no discounting between the time it raises debt and today. Note that today there is no debt). Compute the stock price/share in this case.

Hint #1: You can use what you computed in Q2 as the first part for APV calculation. (Recall, APV = NPV_{OA} + PV(DTS))

Hint #2: Recall, "rule of thumb" (see slides) says use a specific discount rate when the debt policy is such that debt is fixed (which one?). However, try to do sensitivity analysis using both r_A and r_D as discount rates for debt tax shields.

5. How do the values from the APV and WACC approaches compare? If the values are different, what explains the difference?

Hint: Just qualitative discussions will suffice.